SKOGS-TERRASS-ERNA

Design programme

Detailed development plan for
Löjtnanten 1, Västra skogen, Solna

VERSION 2026-10-08 · DRAFT FOR CONSULTATION
SUPERSEDES THE PLANNING BASIS OF 2026-10-05
12 882 M² GFA · 130 HOMES · 6–4 STOREYS

A BLOCK WHERE THE METRO MEETS THE FOREST



00
CONTENTS

02 INTRODUCTION
02 PURPOSE, APPLICATION, ORGANISATION
03 BACKGROUND AND CONCEPT
04 GREEN STRUCTURE
05 NATURAL VALUES AND TARGET SPECIES
06 PROTECTIVE AND COMPENSATORY MEASURES
07 URBAN STRUCTURE AND BRUTALISM
08 ARCHITECTURE AND LANDSCAPE CONCEPT
09 ILLUSTRATIVE PLAN
10 ISOMETRIC VIEWS
12 PUBLIC SPACE
12 JOHAN ENBERGS VÄG AND THE FOREST EDGE
13 THE SUN GATE AND THE ROUTE
14 PRIVATE BLOCK LAND
14 DESIGN PRINCIPLES
15 EAST WING: DESIGN, PLANS, FLATS
18 EAST WING: ISOMETRIC AND JUNCTION
19 MATERIALS AND COLOURS
20 THE FOREST COURTYARD
21 REGULATIONS
22 GREEN MEASURES AND REPLANTING

PURPOSE AND APPLICATION

The design programme for Löjtnanten 1 (Skogsterrasserna) coordinates and secures design quality within the detailed development plan and is proposed as an appendix to the development agreement. Solna stad is responsible for public space; the developer for the block.

The programme guides the building permit assessment under the Planning and Building Act (PBL) as well as design and construction. The qualities, dimensions and materials described are a benchmark: alternative solutions may be accepted if judged to have at least equivalent architectural and ecological quality.

CHANGES SINCE 2026-10-05

– The east wing has four storeys instead of three.
– The east wing is extended from 23.0 m to 32.7 m and joined to the east block at shared slab levels.
– The block: 12 882 m² GFA and 130 homes (+575 m² +7 homes).
– The ecological opening towards Huvudsta forest is concentrated in the Forest Gate, with no garage beneath.

NOTES

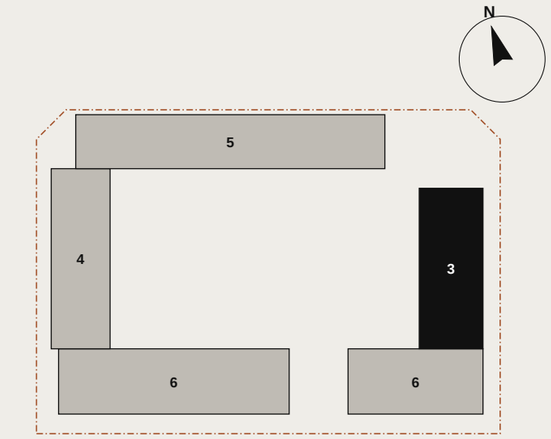
Plans are drawn in the block's local grid: +v points 18.3° east of true north. The north arrow shows true north. All scales refer to A3 prints. Levels are above finished entrance floor (+0.0 ≈ +31.5 in RH 2000, to be verified). Areas are sketch areas to SS 21054:2020 and must be verified. Service shafts are not shown on the plans; they are added at the next stage.

PROJECT ORGANISATION

SOLNA STAD
URBAN PLANNING DEPT. · [NAME]
DEVELOPER
SPG (LAND ALLOCATION 2026) · [NAME]
ARCHITECT
[NAMN]
LANDSCAPE ARCHITECT
[NAMN]
ECOLOGIST / SURVEY
[NAMN]
FIRE, ACOUSTICS, ENERGY
[NAMN]

GRAPHIC REFERENCES

The programme follows the structure of Solna's design programme for the Fanjunkaren block (2024). The graphic form borrows the grid and numbered section tabs from architecture presentation templates, the generous picture editing from AD and the technical precision – numbered details, stated scales, line drawings – from DETAIL. The tone is brutalist: black, concrete grey and one corten accent.



KEY PLAN 1:1250 · BUILDING NUMBERS
6 BLOCKS · 5 NORTH WING · 4 WEST WING · 3 EAST WING (NEW)

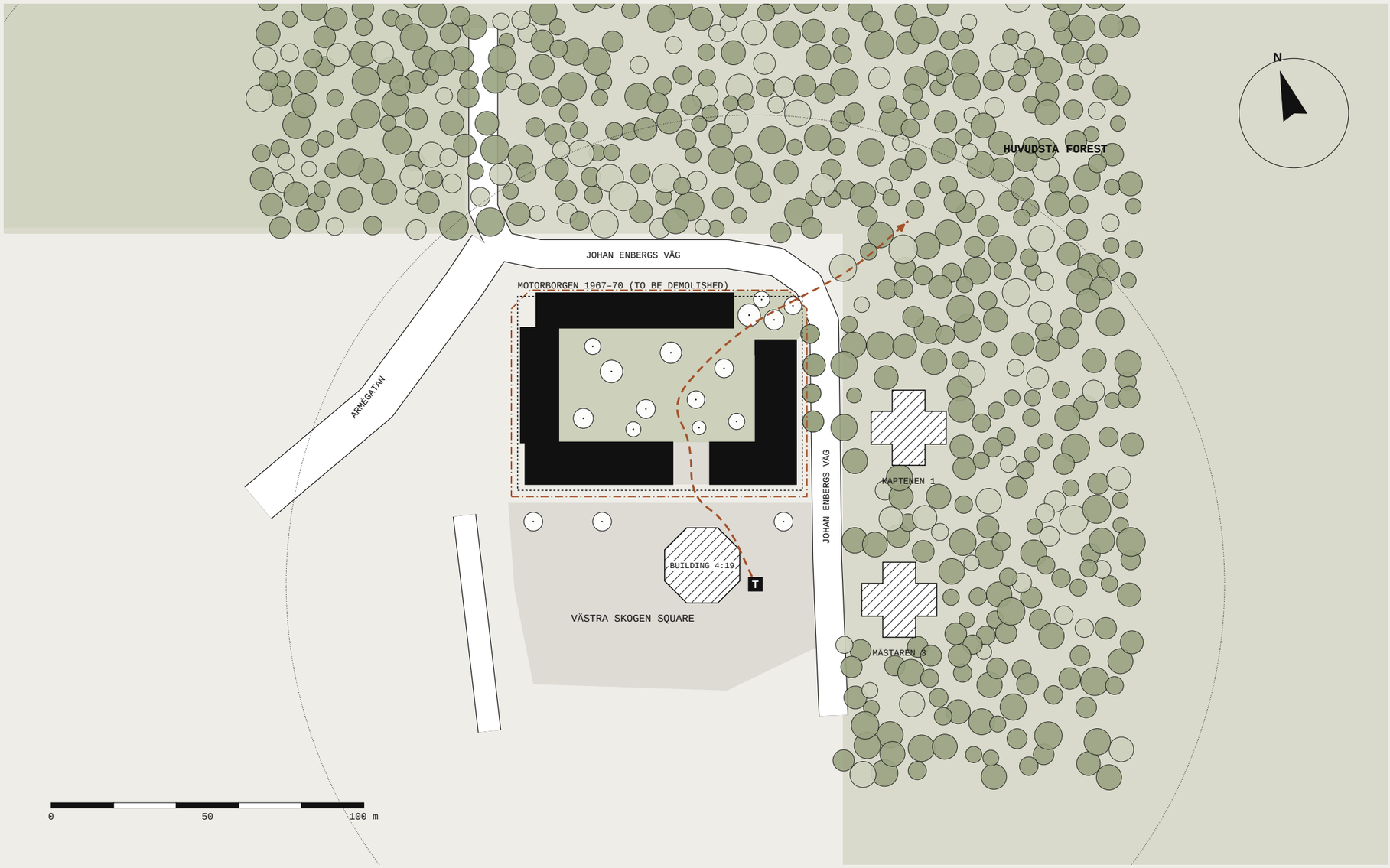
01
BACKGROUND AND CONCEPT

A 1967 car park makes way for a block where the metro meets the forest.

Löjtnanten 1 sits on Västra skogen square, a few minutes from the metro entrance. Today the block is occupied by Motorborgen, a closed car park that turns its back on the square and leaves it dark and empty after office hours. To the east, Huvudsta forest begins on the far side of Johan Enbergs väg.

The proposal replaces the car park with a perimeter block that opens in two directions: towards the square through the twelve-metre Sun Gate, and towards the forest through the Forest Gate in the north-east. Motorborgen's concrete does not disappear. It is sawn into blocks and reused for plinths, retaining walls and habitat piles.

The east wing facing the forest rises to four storeys and is joined to the east block. This closes the block's south-east corner, gives a continuous frontage to Johan Enbergs väg and concentrates the ecological opening in one place, where it can be wide, dark and planted on natural ground.



THE SITE 1:1500 (A3). MOTORBORGEN'S FOOTPRINT DASHED. CIRCLES 150 AND 300 M FROM THE METRO ENTRANCE. THE METRO → SUN GATE → FOREST ROUTE IN CORTEN.

The forest
The pine forest gives Huvudsta and Västra skogen their names.

1967–70
Motorborgen is built as a car park by the square.

1975
The metro reaches Västra skogen.

2026
Land allocation. Motorborgen to be demolished.

2026-10
Design programme. East wing becomes four storeys.

2031
Target: 130 homes occupied.

02
GREEN STRUCTURE

The block becomes a link between the forest's core area and the green routes of the town.

Huvudsta forest and the wooded parts of Västra skogen are identified as core areas in Solna's green plan. Between the core area to the east and the green routes to the south, the site is today a barrier of hard surfacing.

The Forest Gate, the forest courtyard and the Sun Gate form a continuous, multi-layered green structure from the forest edge to the square. The east facade is kept dark as a flight line for bats. No trees at the forest edge are removed for the new east wing; its facade stays on the plot boundary.

KEY

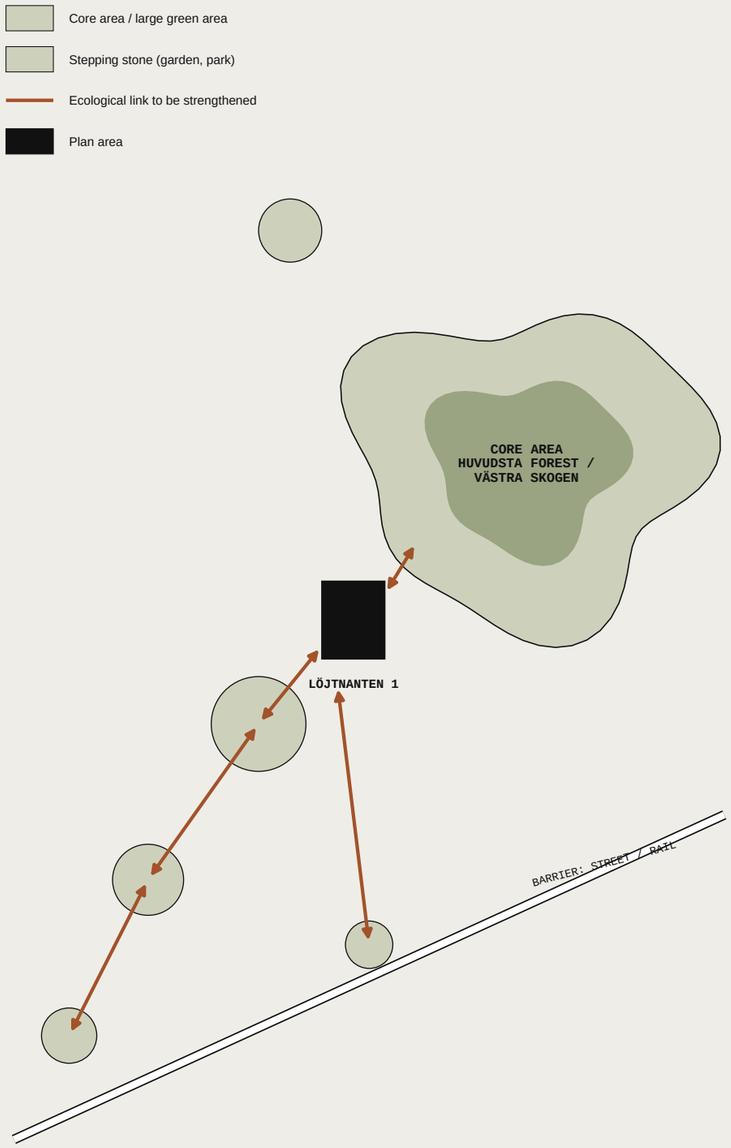
1 The forest edge. The core area's margin is kept and reinforced with a shrub layer.

2 The Forest Gate. Natural ground with no garage deck; at least 8.5 m clear between buildings, 23 m to the street.

3 The Sun Gate. Link to the square for people and pollinators.

4 Dark corridor along the east facade and Johan Enbergs väg, a flight line for bats.

5 Northern link to Huvudsta forest across Johan Enbergs väg.



REGIONAL LINKS. DIAGRAM.
SOURCE: SOLNA GREEN PLAN (2020).

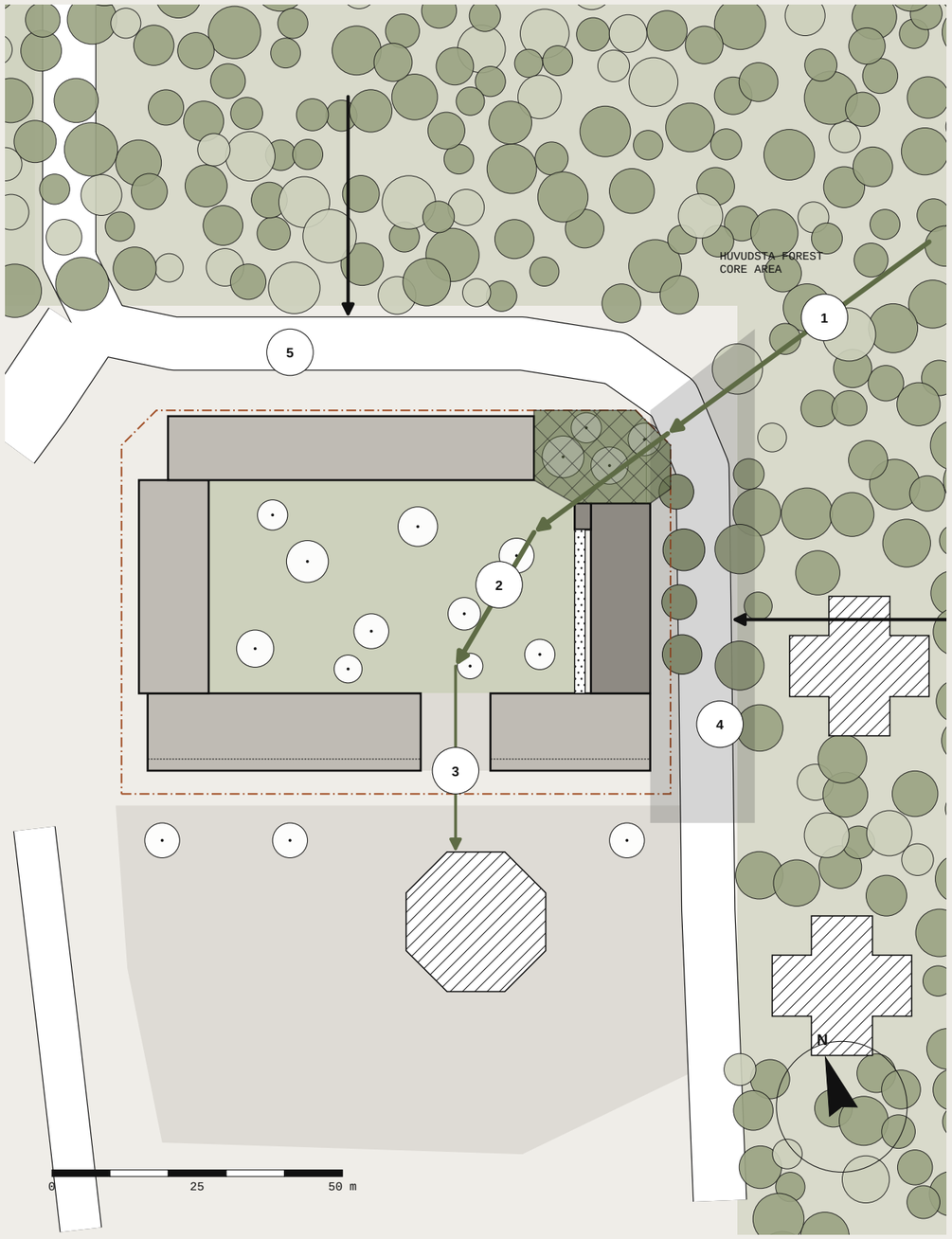


DIAGRAM · NOT TO SCALE

THE BLOCK'S GREEN LINKS 1:1000 (A3). GREY FIELD = DARK CORRIDOR.
GREEN ARROW = NEW DISPERSAL ROUTE THROUGH THE BLOCK.

03
NATURAL VALUES, TARGET SPECIES

Twelve target species determine how the block meets the forest.

A natural values survey to SS 199000 and a species protection assessment must be produced for the plan area and the forest edge. Pending these, the design is based on twelve species typical of the Huvudsta pine forest and its edge, whose needs translate into specific measures.

All wild birds and all bats are protected under the Species Protection Ordinance (ASF). Red List status is given from the 2025 Red List where confirmed; others are checked in Artfakta before consultation.

ASF is amended on 1 November 2026 (SFS 2026:1504). Section references must be checked against the wording in force when the plan is reviewed.

COLOUR CODE

BLACK: BATS · CORTEN: BIRDS
GREY: MAMMALS, AMPHIBIANS · GREEN: INSECTS, FUNGI, PLANTS

01

Northern bat

Eptesicus nilssonii

PROTECTION
Protected, s. 4 a ASF

RED LIST
Check in Artfakta

NEEDS
Dark flight lines, insect-rich edge.

MEASURE
Dark corridor; 3 bat bricks in the east facade, +5.8 m.

02

Soprano pipistrelle

Pipistrellus pygmaeus

PROTECTION
Protected, s. 4 a ASF

RED LIST
Check in Artfakta

NEEDS
Tree lines, insects near damp areas.

MEASURE
3 boxes at the forest edge 4–6 m; rain gardens.

03

Brown long-eared bat

Plecotus auritus

PROTECTION
Protected, s. 4 a ASF

RED LIST
Check in Artfakta

NEEDS
Very light-sensitive; dense vegetation.

MEASURE
The Forest Gate unlit and multi-layered.

04

Common swift

Apus apus

PROTECTION
Protected, s. 4 ASF

RED LIST
EN, Red List 2025

NEEDS
Nests in buildings, free approach.

MEASURE
8 integrated boxes under the eaves, +13.3 m, east and north.

05

Willow tit

Poecile montanus

PROTECTION
Protected, s. 4 ASF

RED LIST
NT

NEEDS
Soft dead wood to excavate its nest.

MEASURE
2–3 m high stumps from felled trees in the Forest Gate.

06

Great spotted woodpecker

Dendrocopos major

PROTECTION
Protected, s. 4 ASF

RED LIST
LC

NEEDS
Old pines, dead wood.

MEASURE
Large pines at the edge released; habitat piles.

07

Hedgehog

Erinaceus europaeus

PROTECTION
Protected (Hunting Act)

RED LIST
VU, Red List 2025

NEEDS
Passages between plots, hibernation in brush.

MEASURE
3 passages 13 × 13 cm in fences and walls; brush piles.

08

Common toad

Bufo bufo

PROTECTION
Protected, s. 6 ASF

RED LIST
—

NEEDS
Damp shelter, no traps.

MEASURE
Rain gardens without high kerbs; stone piles; no open gullies.

09

Pine borer beetle

Nothorhina muricata

PROTECTION
—

RED LIST
NT (2020), check

NEEDS
Sun-exposed old pines.

MEASURE
Release cutting; no work on large pines.

10

Pine conk

Porodaedalea pini

PROTECTION
Indicator species

RED LIST
NT (2020), check

NEEDS
Very old pines.

MEASURE
Root protection zone around large pines during works.

11

Liverleaf

Hepatica nobilis

PROTECTION
Protected, ASF annex 2

RED LIST
LC

NEEDS
Shaded, humus-rich soil.

MEASURE
Any colony moved before works; field layer in the courtyard.

12

Bumblebees, solitary bees

Bombus spp., Andrena spp.

PROTECTION
—

RED LIST
Several red-listed species

NEEDS
Flowers March–October, bare sand.

MEASURE
Meadow roofs, 10 m² sand bed, willow and bird cherry.

04
PROTECTIVE AND COMPENSATORY MEASURES

Measures are numbered as in the isometric. Dimensions and numbers are project targets to be checked against the survey and species assessment; they are not statutory requirements.

01 Swift boxes. 8 integrated under the eaves (+13.3 m) on the east wing's east and north facades, at least 1 m apart, free approach, no windows directly below.

02 Bat bricks. 3 built into the east facade at +5.8 m, never above windows or entrances, no facade lighting.

03 Bat boxes. 3 on trees at the forest edge, 4–6 m high, facing S–SE, clear flight path.

04 Dark corridor. No facade lighting to the east. Street lights ≤ 4 m, full cut-off, ≤ 2,700 K. Target ≤ 1 lux in the canopy.

05 The Forest Gate. Natural ground with no garage deck. Existing soil reused. Pine, oak, rowan, juniper, guelder rose, alder buckthorn, bilberry. No fixed lighting; ≤ 2,200 K ground guide lights at the path.

06 Habitat piles. Dead wood from felled trees and concrete blocks from Motorborgen in gabions, in sun and in shade.

07 The forest courtyard. At least 0.8 m of soil over half the area so pines can grow. Native species only. No pesticides.

08 Biodiverse roof on the east wing. 80–200 mm varied substrate, sand mounds, dead wood, native meadow seed mix.

09 Meadow roof on the north wing; the blocks' roofs combine meadow and solar panels.

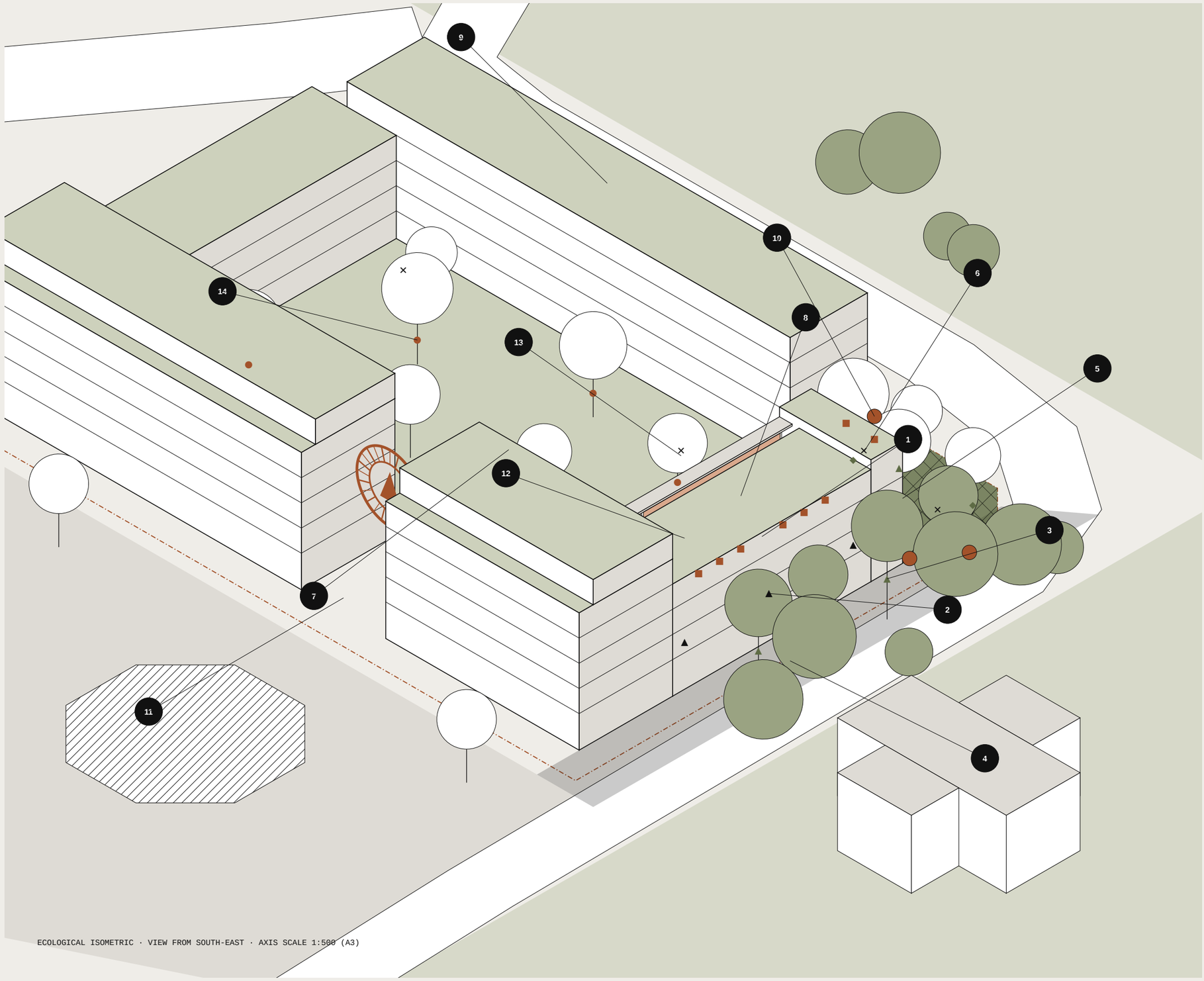
10 Hedgehog passages. 13 × 13 cm openings in fences and walls at the Forest Gate and the plot boundary.

11 Bird-safe glass. Patterned glass, max. 10 cm clear spacing, in the Sun Gate, glazed corners and stair cores.

12 Green walls. Wires with native climbers (hop, honeysuckle, ivy) in the light slot between gallery and facade.

13 Sand bed for wild bees, 10 m² facing south.

14 Bird and wood-mould boxes. 4 tit boxes (Ø 28/32 mm holes) at 2.5–3.5 m facing E–SE; 2 wood-mould boxes in half shade.



ECOLOGICAL ISOMETRIC · VIEW FROM SOUTH-EAST · AXIS SCALE 1:500 (A3)

SYMBOLS: ■ CORTEN = SWIFT BOX · ▲ BLACK = BAT BRICK · ▲ GREEN = BAT BOX · ● = TIT BOX · ▫ = WOOD-MOULD BOX · ▫ = HABITAT PILE · CORTEN DOT ON GROUND = HEDGEHOG PASSAGE. DARK FIELD = DARK CORRIDOR.

05
URBAN STRUCTURE AND BRUTALISM

Concrete carries the history, timber carries the forest.

Västra skogen was built as a modernist district of repeated building types at a large scale, placed in relation to topography and forest. Motorborgen, the Shell building and the metro buildings carry a brutalist tradition: honest use of materials, sculptural volumes and visible structure.

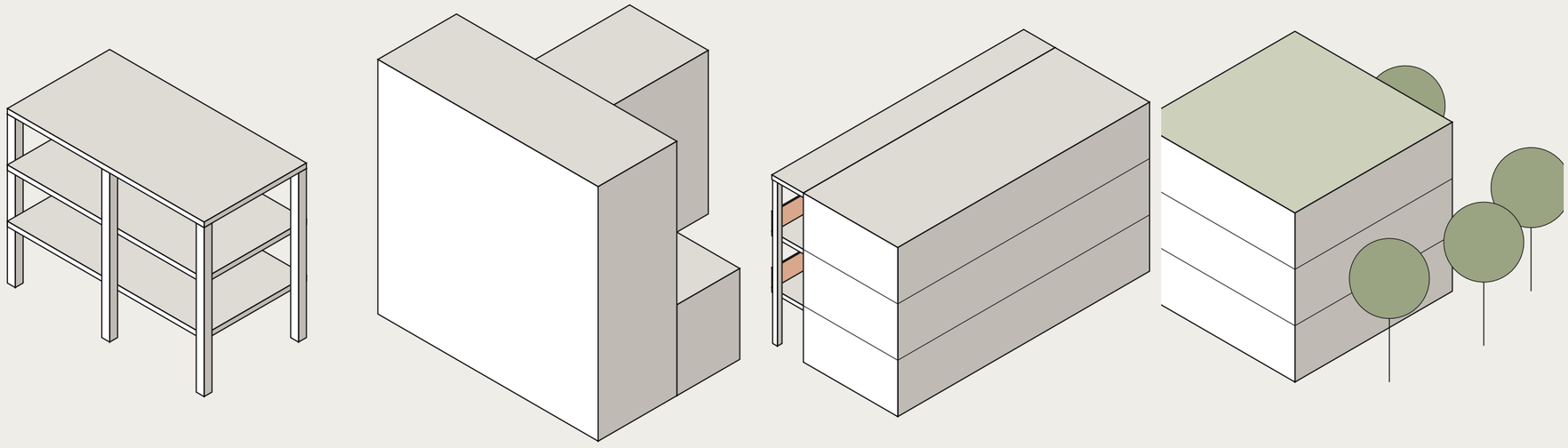
Skogsterrasserna starts from that tradition but reads it through the forest. Most concrete faces the square, most timber faces the forest. The brutalist tone lies in the frame, in the gallery and in the exposed slab edges, not in a closed face to the street.

HEIGHTS

BLOCKS 6 STOREYS · +20.0
NORTH WING 5 STOREYS · +15.8
WEST WING 4 STOREYS · +12.7
EAST WING 4 STOREYS · +13.8 (+14.3 PARAPET)
KAPTENEN 1 · 3 STOREYS

REFERENCES

MOTORBORGEN, SOLNA (1967–70)
SHELL BUILDING / THE BLOCK, SOLNA
METRO BUILDING, VÄSTRA SKOGEN
PARK HILL, SHEFFIELD
BARBICAN ESTATE, LONDON



01

Honest structure
Frame of reused and cast-in-place concrete. Slab edges and columns are expressed on the facade.

02

Sculptural volumes
Height steps 6–5–4–4 from square to forest. The volumes read as clear blocks.

03

The gallery as a street
The east wing's free-standing gallery is the building's shared room and links to the block's stair core.

04

Green in the concrete
Climbers, biodiverse roofs and the forest courtyard turn concrete into a carrier of life.

APPROACH

Every material shows what it does. Concrete where it carries, timber where it protects, corten where it meets weather and light. No finishes that imitate other materials.

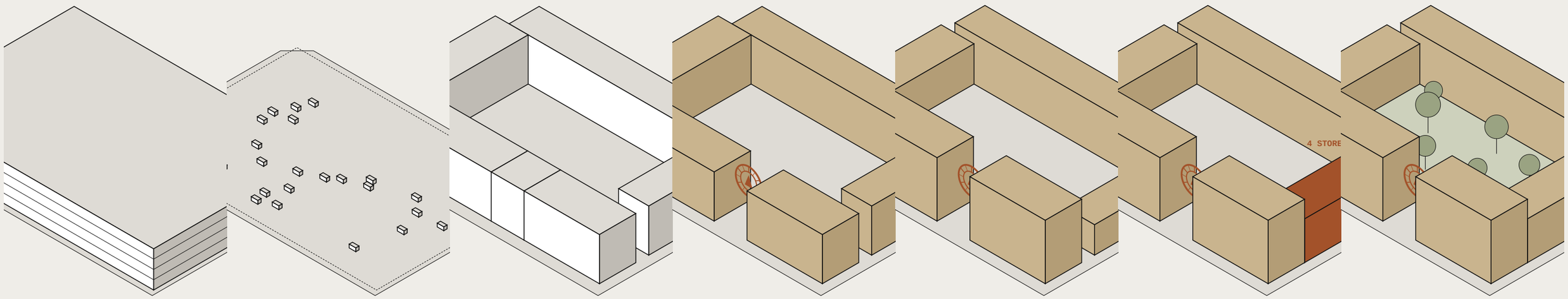
Reuse is part of the expression: the sawn blocks from Motorborgen show their old surfaces in plinths and walls.

WHAT IS NOT DONE

No projecting balconies towards the forest, since they spill light into the edge. No glass balustrades without markings. No facade lighting to the east. No surfaces that need pesticides.

06
CONCEPT

Seven steps from car park to forest block. Step 6 is the change introduced by this programme.



1 Today
Motorborgen 1967–70. A closed car park.

2 Demolish and reuse
Concrete sawn into blocks: plinths, walls, habitat piles.

3 Perimeter block
The block closes around a courtyard and gives the square a facade.

4 Open the Sun Gate
A 12 m gate brings in sun, people and views.

5 Step down to the forest
Six storeys at the square, lower towards the forest.

6 East wing: four storeys
Extended to 32.7 m and joined to the east block.

7 Let the forest in
Forest courtyard and Forest Gate free of cars.

ARCHITECTURE CONCEPT

– Heights and placement follow an established logic: tallest at the square and metro, lower towards the forest.
– All flats from three rooms are dual-aspect, with light from the square or forest and contact with the quiet courtyard.
– Towards the square an urban side with 4.5 m high commercial spaces; towards the forest a lower building in larch and concrete.
– The east wing's gallery gives every flat two escape routes and connects the building to the block.

LANDSCAPE CONCEPT

– The design strengthens the ecological links between Huvudsta forest and the town's green routes.
– The courtyard is forest floor: pines, dwarf shrubs, moss, boulders and play in natural materials.
– Felled trees and sawn concrete become habitat piles, nest boxes and site-made furniture.
– Lighting is planned for darkness: as little, as low and as warm as possible towards the forest.

07
ILLUSTRATIVE PLAN

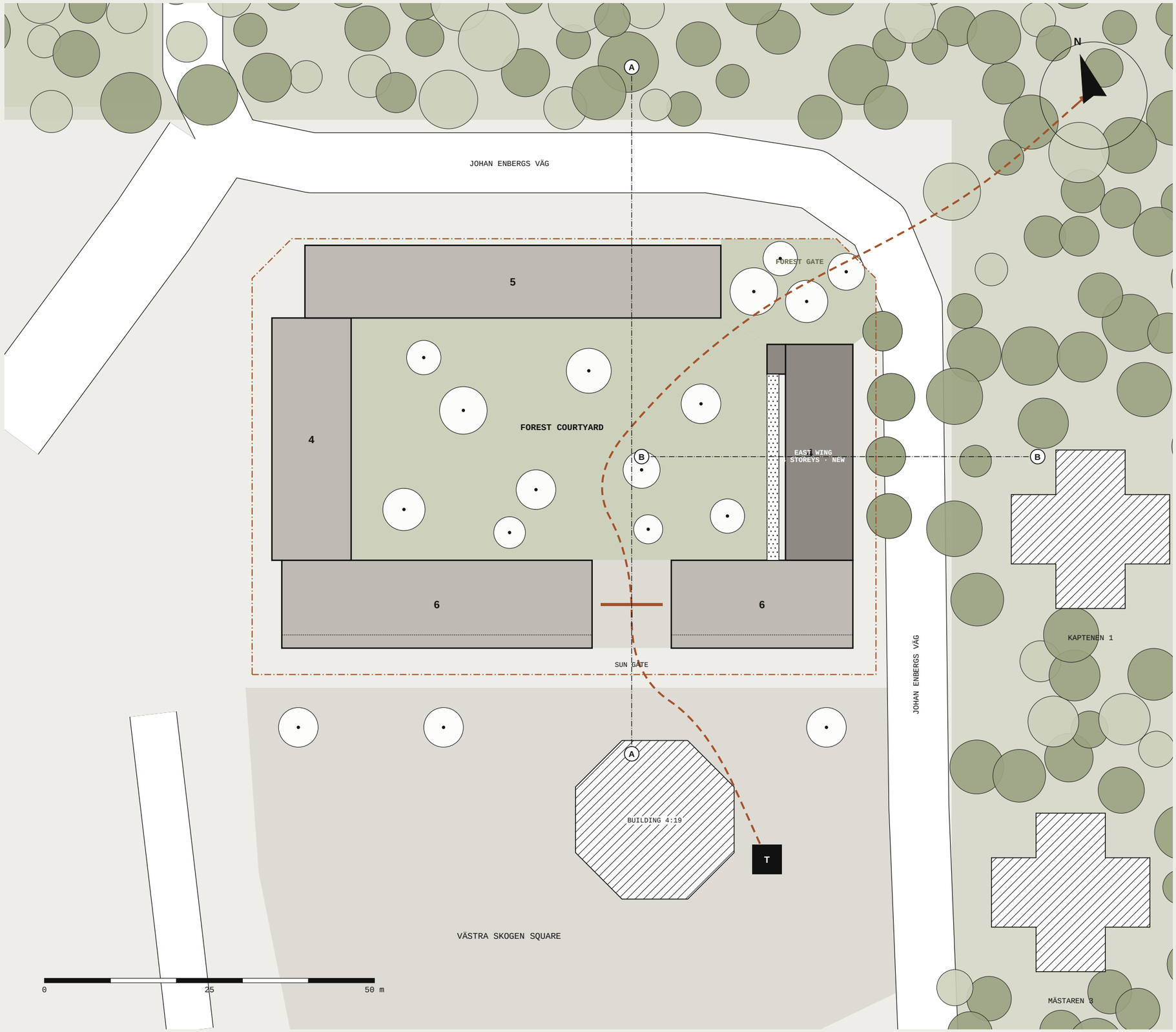
Scale 1:600 (A3)

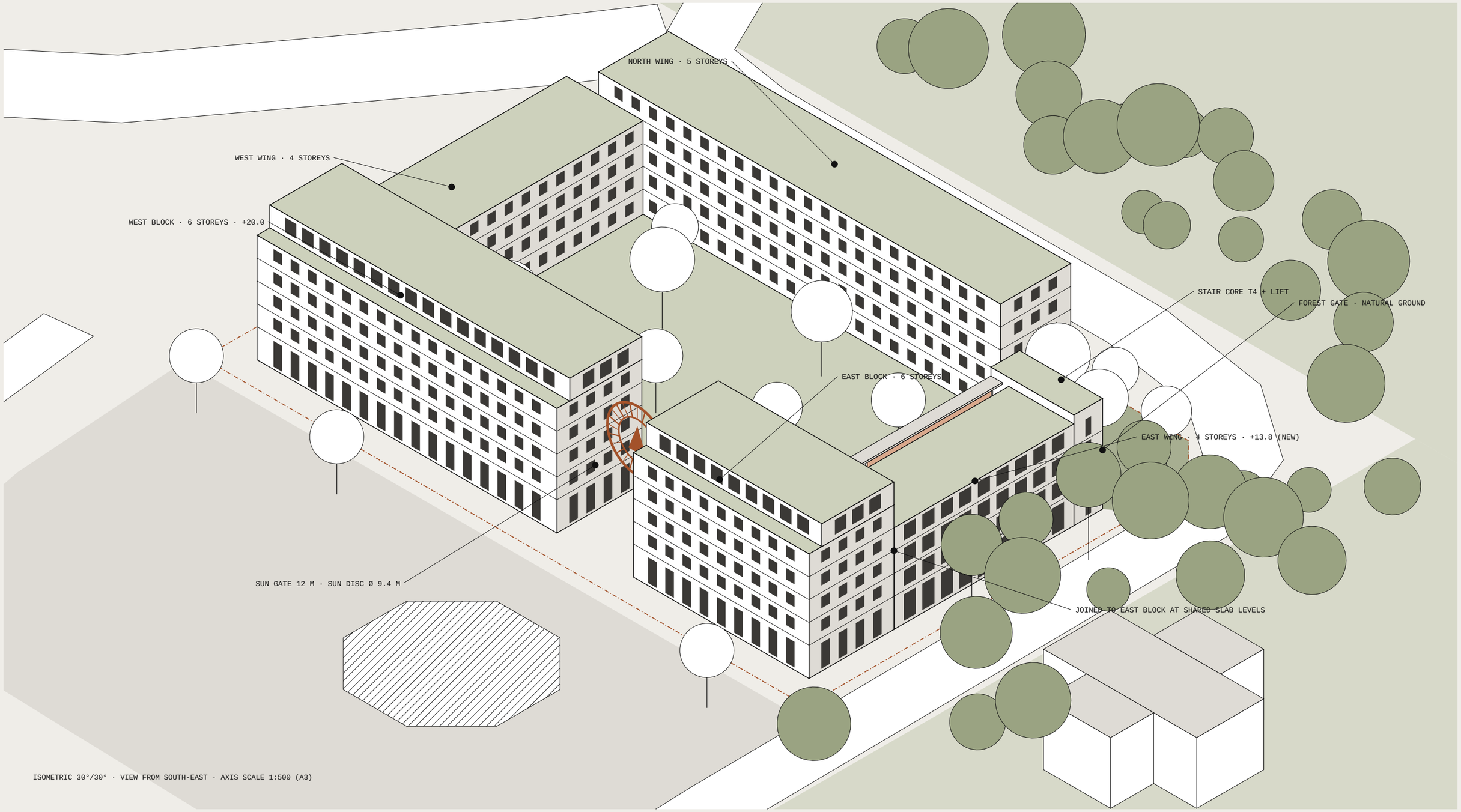
The block consists of five buildings around the forest courtyard. The east wing follows Johan Enbergs väg and meets the east block without a gap. The Forest Gate in the north-east is the block's only opening to the forest and therefore its most important green space.

KEY

- 6 · West and east blocks, 6 storeys. 4.5 m commercial floor to the square.
- 5 · North wing, 5 storeys.
- 4 · West wing, 4 storeys.
- 4 · East wing, 4 storeys, new. Dotted = gallery.
- Corten bar = sun disc in the Sun Gate.
- Dashed corten = metro → forest route.
- White circles = new trees. Green = existing (schematic, survey required).
- A–A, B–B = sections, pp. 12–13.

Surroundings drawn from Solna's base map, accuracy about ±3 m. Existing building volumes simplified.





08
ISOMETRIC

True isometric, 30°/30°. All three axes share one scale, 1:500 when printed at A3, so lengths can be measured directly along the building edges. Floor lines and window rhythm are drawn from the plans.

The east wing joins the east block at the same slab levels (+4.5 / +7.6 / +10.7). The gallery facing the courtyard ends in the block's stair core T3 to the south and in the new stair core T4 with a lift to the north.

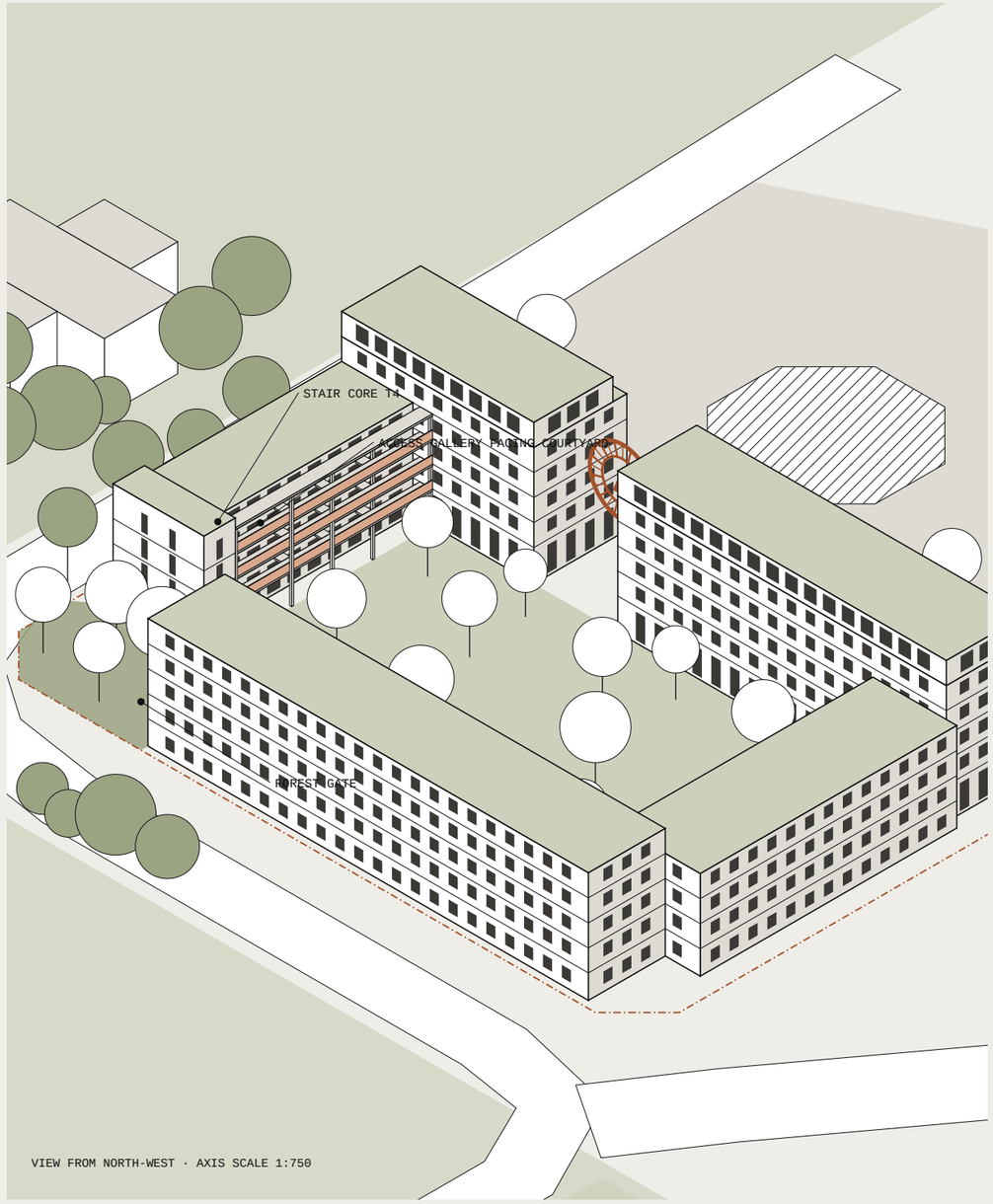
09
THE CHANGE

From three to four storeys, from free-standing to joined.

The east wing is extended from 23.0 m to 32.7 m. It now starts at the east block (v 17.3) and ends at the Forest Gate (v 50.0). GFA rises from 759 m² to 1 334 m² homes from 6 to 13.

The connection creates a continuous frontage to Johan Enbergs väg and shields the courtyard from traffic noise. At the same time the gap between buildings 3 and 6 disappears. This is compensated by making the Forest Gate wider, on natural ground and kept dark.

Four storeys with a 4.5 m ground floor give an overall height of 14.3 m above entrance level. This must fit within the plan's height regulation, see p. 21.



ISOMETRIC FROM THE NORTH-WEST, AXIS SCALE 1:750. THE GALLERY AND STAIR CORE T4 FACING THE COURTYARD.



COMPARISON 1:500. LEFT THE PLANNING BASIS, RIGHT THIS PROGRAMME. CORTEN = CONNECTION.

12 882	130	+575	+7	32.7	14.3
m² GFA in the block	homes	m² GFA east wing	homes east wing	m east wing length	m east wing overall height

10
JOHAN ENBERGS VÄG AND THE FOREST EDGE

PUBLIC SPACE

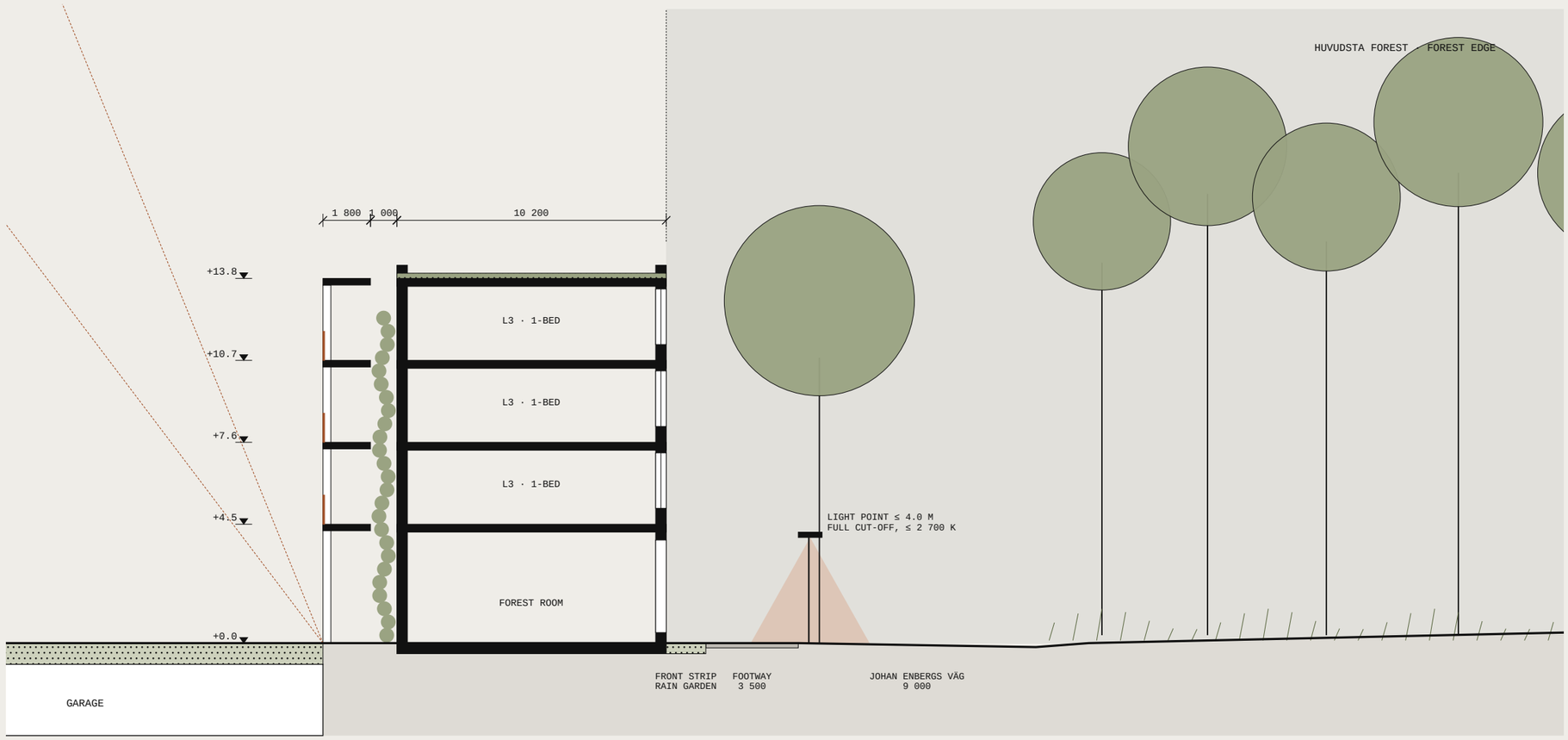
Johan Enbergs väg is the boundary between the block and Huvudsta forest. The street keeps its 9 m width but gains a front strip with rain gardens along the east wing and a 3.5 m footway.

The forest edge on the east side of the street is retained. Old pines are released so they become sunlit, which benefits wood-living insects such as the pine borer. A shrub layer of juniper, guelder rose and alder buckthorn is built up towards the street.

LIGHTING

Bats hunt along dark edges. Lighting is therefore planned for darkness: as few light points as possible, as low as possible and in warm colour. Fittings have flat glass and closed housings so no light goes upward.

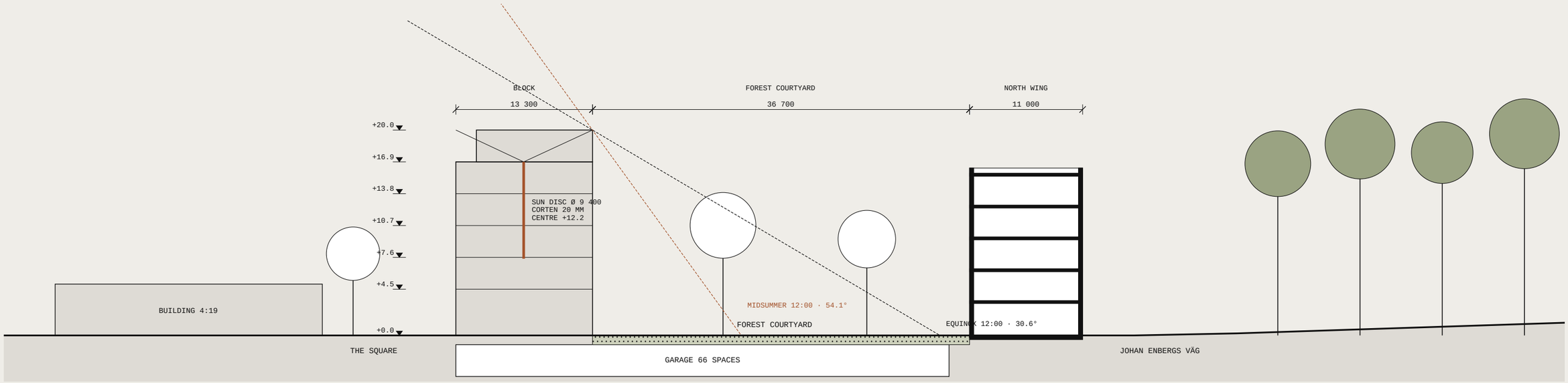
Stormwater is managed in rain gardens along the facade and in the street plantings. In cloudbursts water runs on the surface towards the courtyard and on to the low point.



SECTION B–B, WEST → EAST THROUGH THE EAST WING, JOHAN ENBERGS VÄG AND THE FOREST EDGE. SCALE 1:200 (A3).

LIGHTING PRINCIPLES (PROJECT TARGETS)

Zone	Height	Colour temp.	Control
Square, Sun Gate	≤ 6 m	3 000 K	Dimmed 30 % 23–05
Courtyard, gallery	bollards ≤ 1 m	2 700 K	Presence-controlled
Johan Enbergs väg	≤ 4 m	≤ 2 700 K	Full cut-off, 0 % upward light
Forest Gate	in ground	≤ 2 200 K	Guide lights at path only
East facade	—	—	No facade lighting
East entrances	2,4 m	2 700 K	Downlight, motion sensor



SECTION A–A, SOUTH → NORTH THROUGH THE SUN GATE AND THE COURTYARD. SCALE 1:400 (A3). MIDDAY SUN ANGLES FOR SOLNA, 59.4° N.

11
THE SUN GATE AND THE ROUTE

The Sun Gate is a 12 m wide public passage through the blocks, secured by a plan regulation. The route runs from the metro entrance, through the gate and courtyard, on through the Forest Gate to Huvudsta forest.

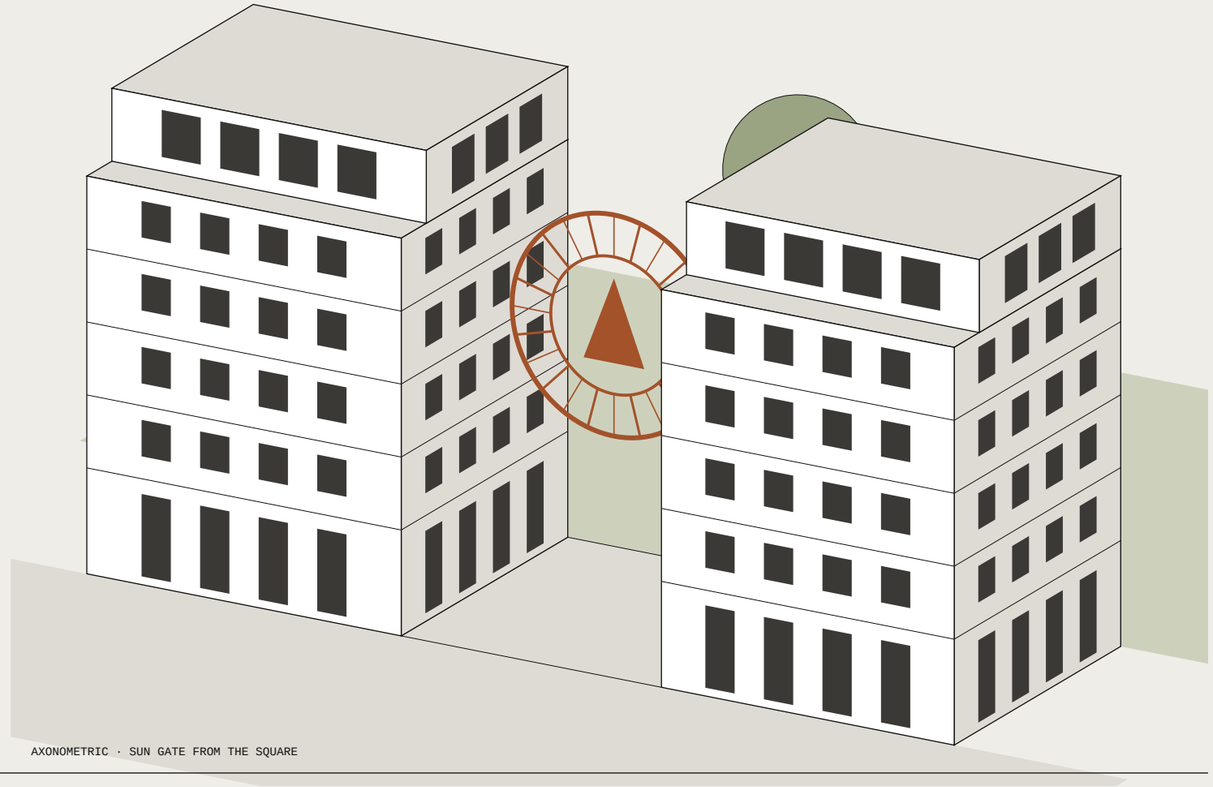
The sun disc, Ø 9.4 m in 20 mm corten steel, hangs with its centre at +12.2 m from four steel cables between the blocks. It shows a sun with 24 rays and a pine at its centre. When the sun is due south it draws its pattern on the square.

MATERIALS SHIFT FROM TOWN TO FOREST

– Square: granite setts.
– Gate: gravel and stone dust, a brass sun compass set into the paving.
– Courtyard: moss, dwarf shrubs, boulders of sawn concrete.

SUN AND SHADE

At the equinox the midday sun is 30.6° above the horizon and the block shades the southern part of the courtyard. The courtyard is therefore designed as forest floor with shade-tolerant species. At midsummer, 54.1°, the sun reaches its northern half.



AXONOMETRIC · SUN GATE FROM THE SQUARE

12
DESIGN PRINCIPLES

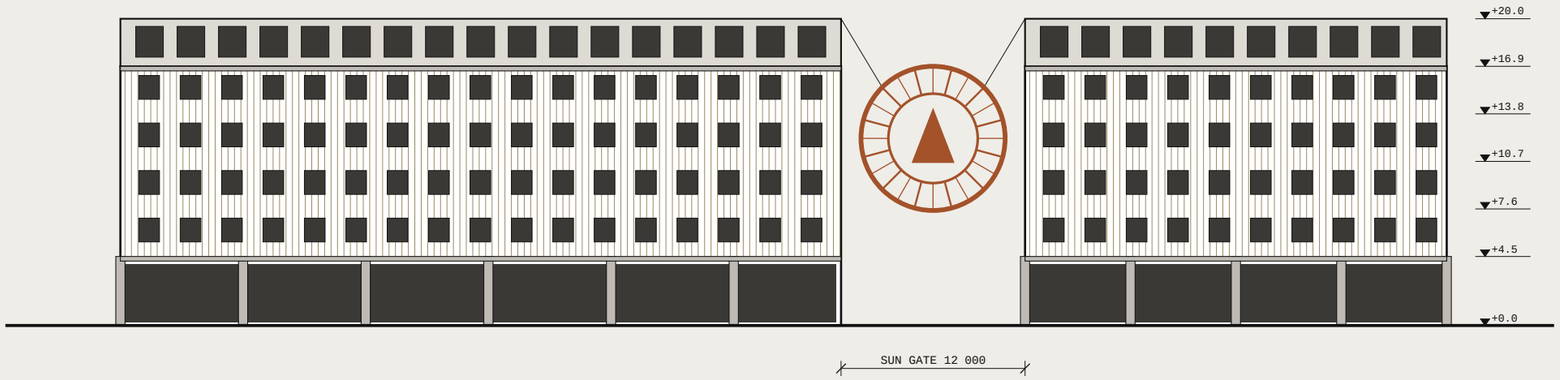
BLOCK LAND

Most concrete faces the square; only timber and concrete face the forest.

To the square. The ground floor is 4.5 m high and fully glazed, with space for a café, bakery and services. Concrete columns every eight metres carry a plinth band. The facades above are clad in vertical heat-treated pine battens in an irregular rhythm, like trunks in a pine forest. The top floor is set back with shared roof terraces facing south.

To Johan Enbergs väg. The block's gable, the east wing and the north wing's gable form one frontage with three heights: 20.0, 14.3 and 15.8 m. The Forest Gate breaks the frontage and shows the forest.

To the courtyard. White-pigmented pine reflects daylight into the shaded courtyard. Balconies have slender steel brackets and timber decks.



FACADE TO THE SQUARE, SOUTH. SCALE 1:400 (A3).



FACADE TO JOHAN ENBERGS VÄG, EAST. SCALE 1:400 (A3). THE EAST WING IN THE CENTRE, THE EAST BLOCK ON THE LEFT, THE NORTH WING BEHIND THE FOREST GATE.

13
EAST WING DESIGN

A building in concrete and larch that meets the forest without shining into it.

Structure. Concrete slab edges and columns are expressed on the facade and divide it into a clear grid. The columns stand at the party walls, so the grid reveals how the building is divided.

Infill. Vertical untreated larch boarding that weathers to grey. Towards the forest the windows sit in deep reveals and the loggias are recessed into the volume. No balconies project towards the edge.

The gallery. On the courtyard side a free-standing gallery stands 1.0 m from the facade, with bridges at each entrance. The gap becomes a green wall of climbers and gives the courtyard-facing bedrooms both light and privacy. Balustrades in perforated corten plate, 1.1 m high.

Eaves. Swift boxes are built into the eaves and bat bricks into the east facade. The roof is a biodiverse roof.



EAST FACADE (JOHAN ENBERGS VÄG). SCALE 1:200 (A3).



WEST FACADE (COURTYARD) WITH GALLERY. SCALE 1:200 (A3). MIRRORED: NORTH TO THE LEFT.

14
EAST WING PLANS

Scale 1:200 (A3)

Level 0, +0.0. 4.5 m high, in line with the blocks' commercial floor. Bike store, through entrance hall, waste room to the street, the Forest Room and a studio flat.

Levels 1–3, +4.5 / +7.6 / +10.7. Four dual-aspect flats per floor, reached from the gallery.

Escape. The gallery leads two ways: south to the block's stair core T3 and north to stair core T4. Every flat has two independent escape routes.

Accessibility. Lifts in T4 and T3. Dashed circles show Ø 1.3 m turning space in bathrooms and Ø 1.5 m in the stair hall. Doors 0.9 m.

Not shown. Service shafts are added at the next stage.

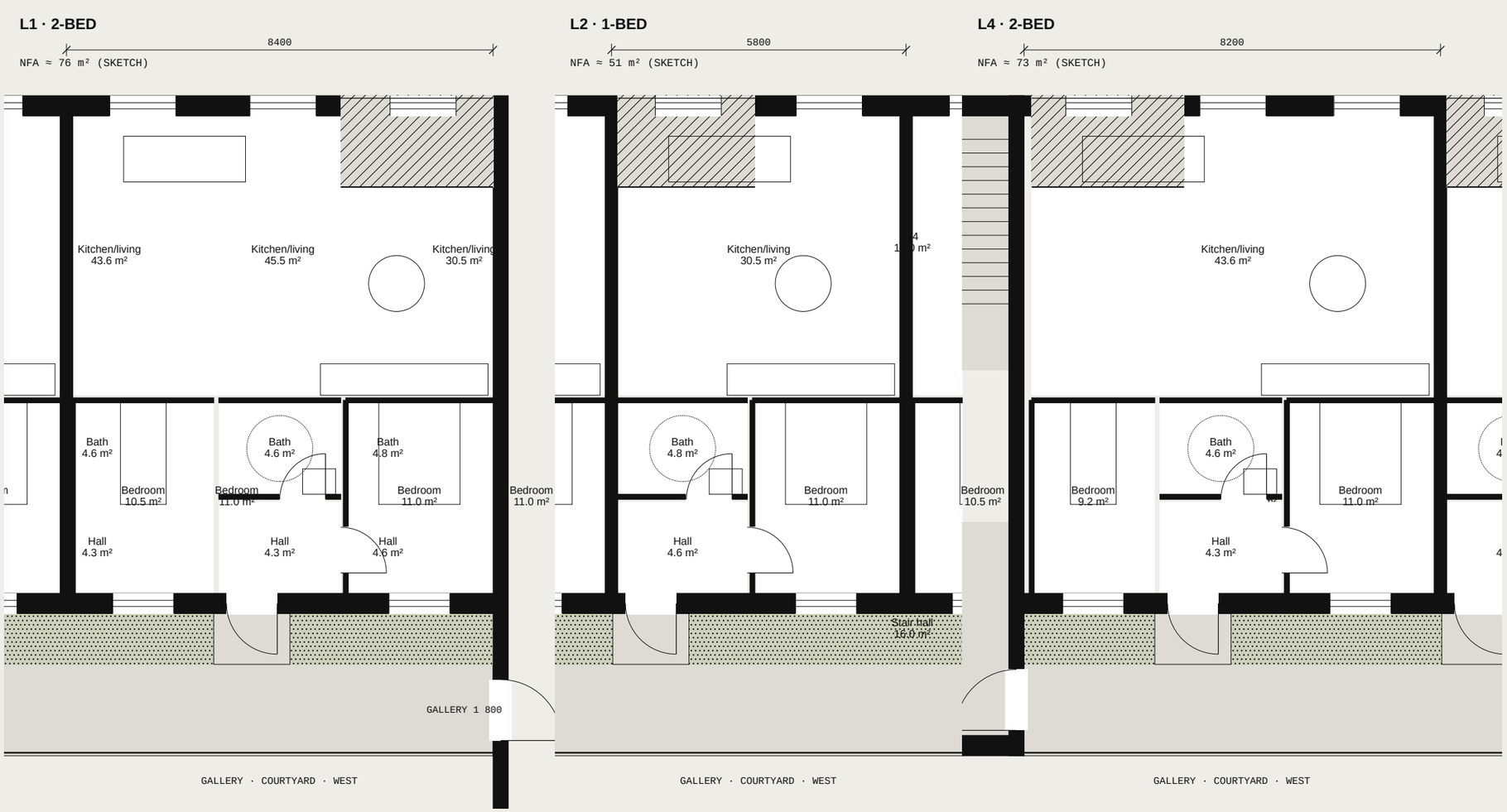


15
EAST WING FLATS

Dual-aspect flats with bedrooms to the courtyard and living rooms to the forest.

The western zone, 3.8 m, holds bedrooms, hall and bathroom. The eastern zone, 5.6 m, is one continuous kitchen and living room facing the forest, with a recessed loggia. Flats are reached from the gallery across a bridge.

Areas are sketch areas measured inside external walls and must be verified to SS 21054:2020.



FLAT PLANS 1:100 (A3), LEVEL 1. FOREST AT THE TOP.

FLAT MIX, EAST WING

Flat	Type	Width	NFA sketch	No.	Level
L1	2-bed	8.4 m	76 m²	3	1–3
L2	1-bed	5.8 m	51 m²	3	1–3
L3	1-bed	5.8 m	51 m²	3	1–3
L4	2-bed	8.2 m	73 m²	3	1–3
A	1-bed	6.5 m	57 m²	1	0
Total				**13**	

16
EAST WING ISOMETRIC

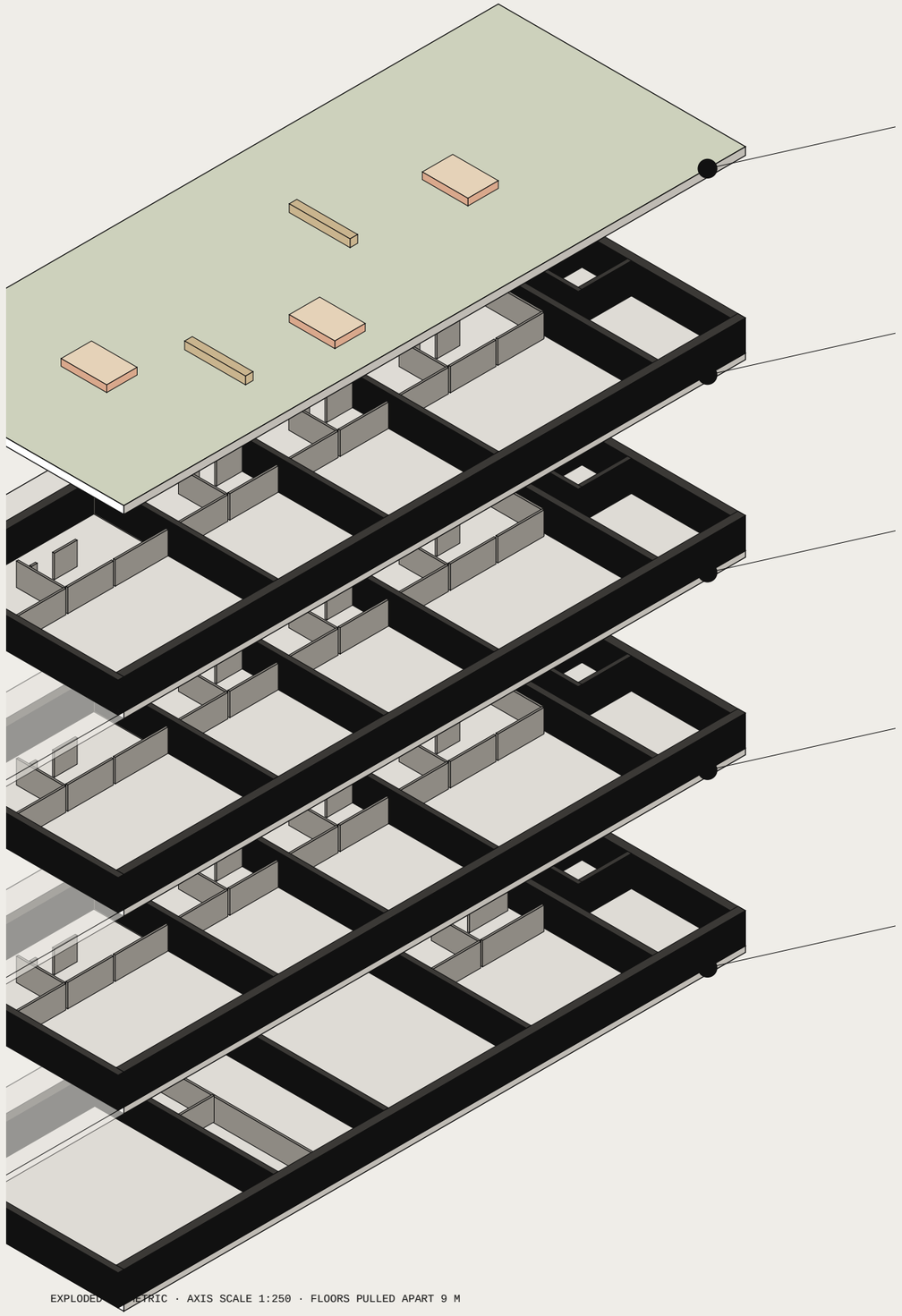
Four floors and a roof, pulled apart so each level can be read.

The exploded isometric shows the walls 1.6 m high above each slab. Black walls are load-bearing or separating; grey walls are light partitions.

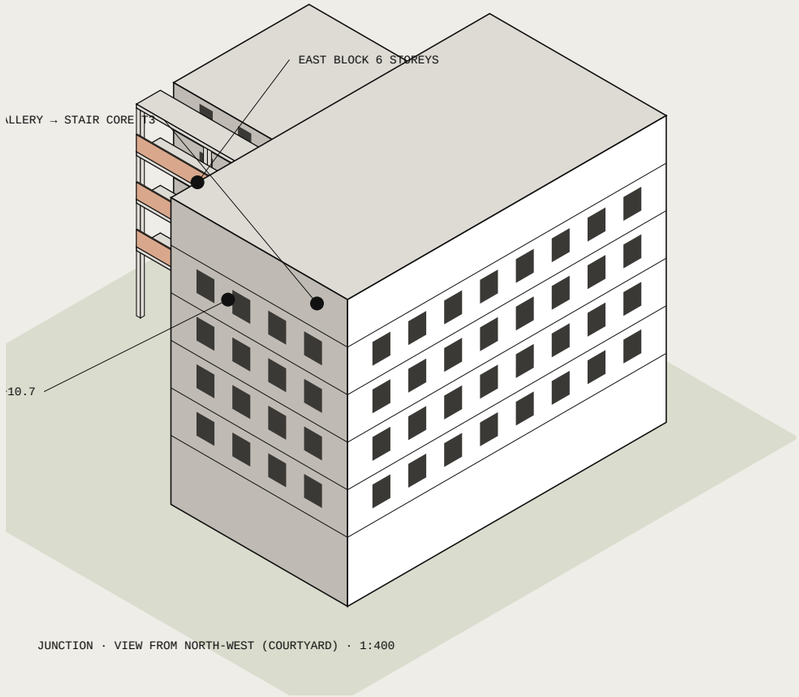
THE JUNCTION

The east block and the east wing share slab levels on levels 1–3. The gallery runs straight into the block's stair core T3 through a fire separation (door in the block's facade line). The structures are separated by a movement joint so the buildings can move independently.

The 4.5 m ground floor is needed for the slabs to align. It gives the Forest Room and the studio a ceiling height suited to shared uses.

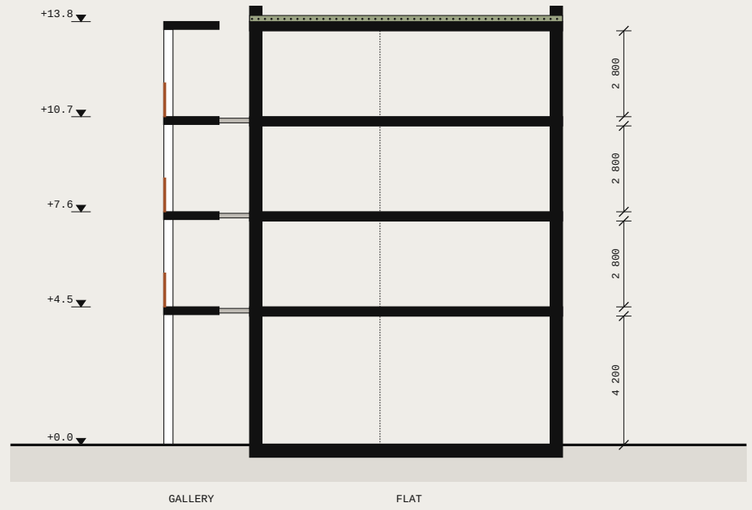


EXPLODED ISOMETRIC · AXIS SCALE 1:250 · FLOORS PULLED APART 9 M



JUNCTION · VIEW FROM NORTH-WEST (COURTYARD) · 1:400

THE JUNCTION SEEN FROM THE COURTYARD.



SECTION C–C THROUGH THE EAST WING. SCALE 1:200 (A3).

17
MATERIALS AND COLOURS

Materials shift from town to forest.

The palette is short: concrete, timber, corten and plants. The concrete has two origins, cast in place and reused, and both are left untreated. The timber is larch and pine from Swedish forests. Corten is used where metal meets weather and light.

The colours relate to the area's concrete, brick and bronze-anodised aluminium. No biocide-treated materials are used in facades facing the forest.



01

Board-marked cast concrete

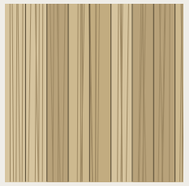
Columns, slab edges



02

Reused concrete

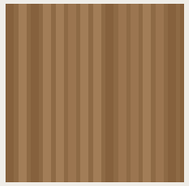
Plinths, walls from Motorborgen



03

Untreated larch

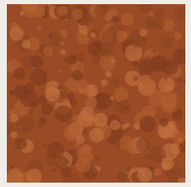
East wing, vertical boards



04

Heat-treated pine

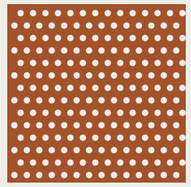
Square and street facades



05

Corten steel

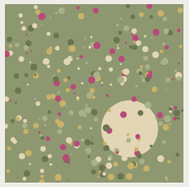
The sun disc



06

Perforated corten plate

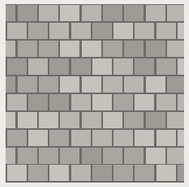
Gallery balustrades



07

Biodiverse roof, meadow

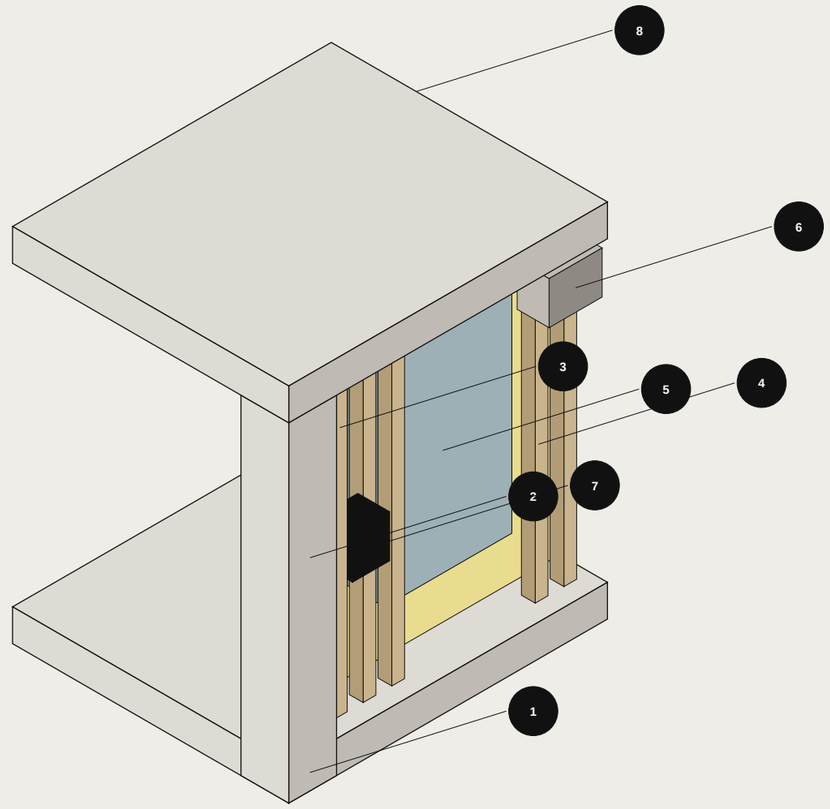
Roofs



08

Granite setts

The square



FACADE FRAGMENT EAST WING TO FOREST · ISOMETRIC · AXIS SCALE 1:50

FACADE FRAGMENT, EAST WING TO THE FOREST, 1:50

1 Slab edge, board-marked cast concrete

2 Column 450 × 450, reused concrete

3 Vertical larch boards 28 × 120 on battens, air gap

4 Insulation in timber stud wall (U-value per energy calculation, BFS 2026:9)

5 Timber/aluminium window, deep reveal

6 Swift box built into the eaves

7 Bat brick, built in

8 Floor slab, concrete

18
THE FOREST COURTYARD

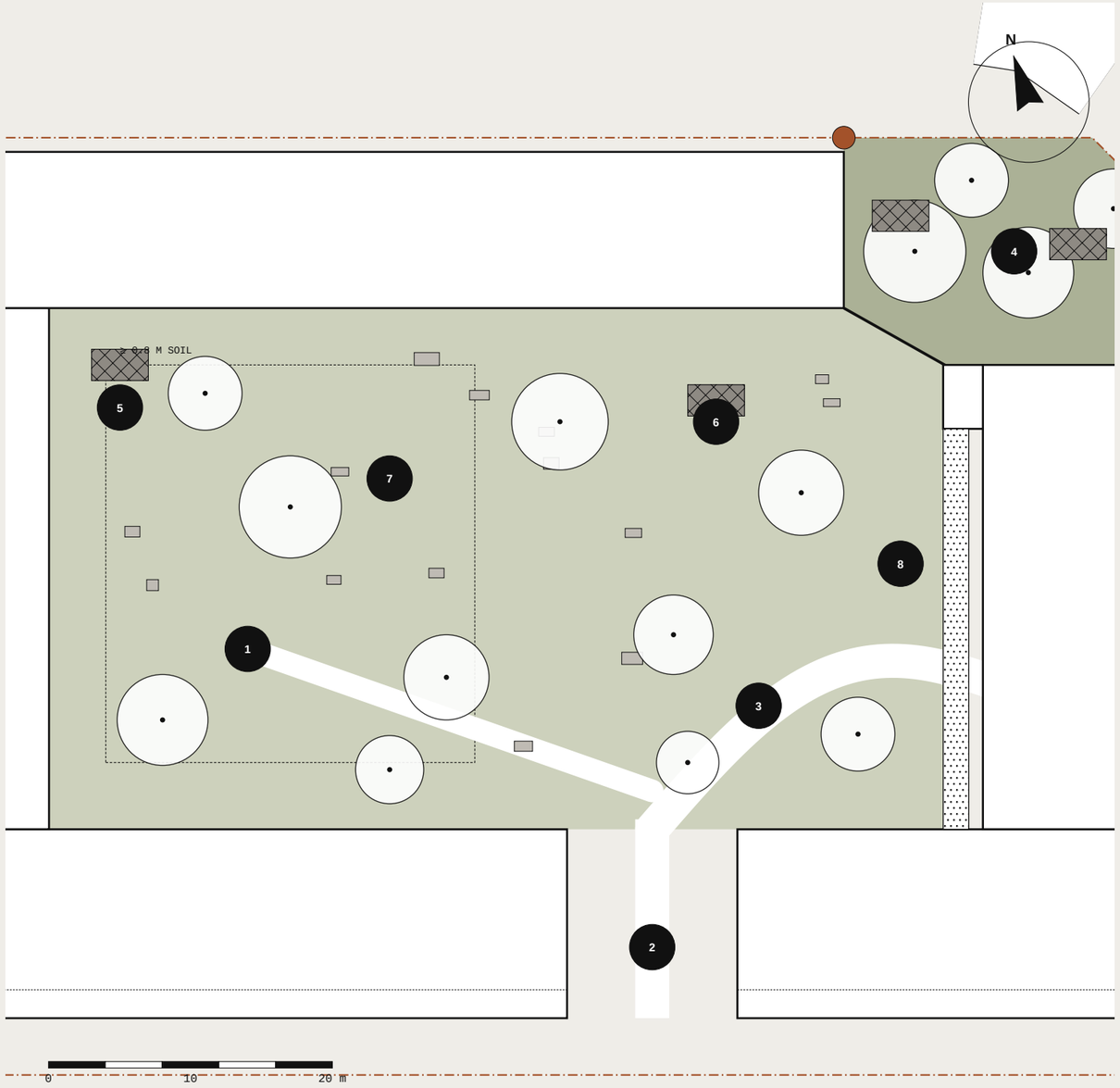
A piece of forest floor on the garage roof.

The courtyard sits on the garage deck. At least half its area has 0.8 m of soil so pines can grow; the rest has 0.4–0.6 m for dwarf shrubs, moss and low bushes. Retaining walls of sawn concrete blocks from Motorborgen take up the levels.

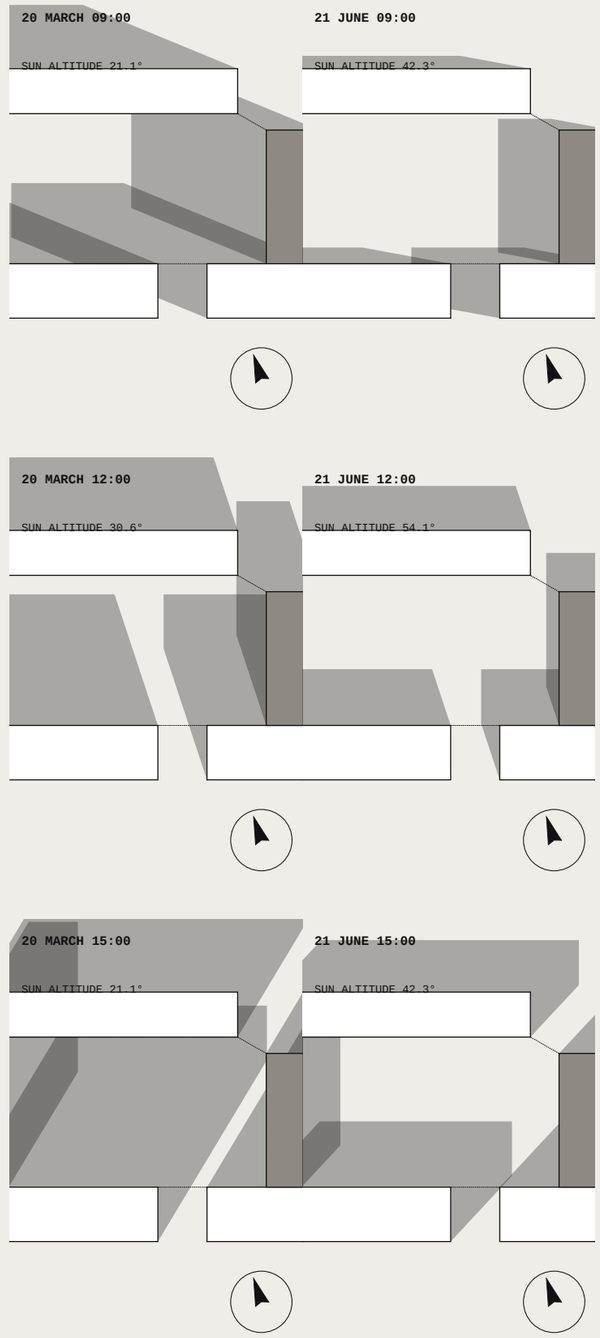
The courtyard is car-free. Drop-off is from the streets. Play, growing and gathering fit in the forest rooms between the trees. Lighting is low and presence-controlled.

KEY

1 Gravel path recalling the car park's ramps
2 The Sun Gate
3 Play in natural materials: logs, boulders, sand
4 The Forest Gate on natural ground
5 Habitat pile of brush and trunks
6 Sand bed for wild bees
7 Pines in at least 0.8 m soil
8 Light slot with climbers by the gallery



PARTIAL PLAN OF THE FOREST COURTYARD 1:400 (A3).



SHADOW STUDY 1:1500. SOLAR TIME, SOLNA 59.4° N. EAST WING GREY. THE EAST WING'S FOURTH STOREY ADDS NO NEW SHADE ON THE COURTYARD BEFORE 15:00 AT EQUINOX.

19
REGULATIONS

Summary as of 2026-10-08.

Since 1 July 2026 only Boverket's new building rules apply to new cases. They are split into separate regulations per requirement area and are performance-based. The energy rules in BBR were replaced on 1 October 2026 by BFS 2026:9.

LOCAL CONDITIONS, SOLNA

Height. The east wing's overall height is 14.3 m above entrance level, about +45.8 in RH 2000, and the lift overrun reaches 15.5 m. Because of the 4.5 m ground floor, height should be regulated in metres above datum, not in number of storeys, so a permit can be granted without deviation.

The detailed plan should also secure the Sun Gate as a public passage, the Forest Gate as land that may not be built on or under, and design provisions for the gallery and facade materials.

Other. Public space follows Solna's design programme for public environments. Parking (66 car spaces, 420 cycle spaces) and stormwater are checked against the city's current guidelines. Solna's building committee grants permits; the Stockholm County Administrative Board decides on any species protection derogation.

References must be checked against the wording in force at permit application.

Area	Regulation (as of 2026-10-08)	Application in the project
Planning, permits	Planning and Building Act (2010:900), amended 2025-12-01 (new ch. 9); Planning and Building Ordinance (2011:338)	Detailed plan for Löjtnanten 1. Programme appended to the development agreement. Regulations per Boverket's catalogue of plan provisions.
Building rules	Boverket's new building rules apply in full to new cases since 2026-07-01. BBR and EKS are repealed.	Design follows the BFS 2024 series and BFS 2026:9.
Care, demolition	BFS 2024:4	Demolition plan and material inventory for Motorborgen; concrete is reused.
Structure	BFS 2024:6	Reused concrete tested and documented. Movement joint at the junction.
Fire	BFS 2024:7	Two independent escape routes via the gallery (T3, T4). Fire separation at the junction. Fire safety report.
Health, environment	BFS 2024:8	Daylight from two sides; radon measurement; moisture safety design.
Safety in use	BFS 2024:9	Gallery balustrades 1.1 m without climbable parts; slip-resistant surfaces.
Noise	BFS 2024:10; Ordinance (2015:216) on traffic noise at housing	Noise study. Dual-aspect flats with a quiet side to the courtyard.
Housing design	BFS 2024:11	Dual-aspect flats, storage, room sizes per plans.
Accessibility	BFS 2024:12	Lifts in T3 and T4, level entrances, turning spaces, 0.9 m doors.
Plots	BFS 2024:13	Open space in the courtyard and Forest Gate; drop-off near entrances.
Energy	BFS 2026:9, in force 2026-10-01, transition to 2027-09-30	Nearly zero-energy building; energy calculation at next stage.
Climate	Act (2021:787) on climate declarations for buildings	Reuse of Motorborgen's concrete is reported.
Species	Environmental Code ch. 12 s. 6; Species Protection Ordinance (2007:845), amended to SFS 2026:1504, new wording from 2026-11-01	Species assessment; time windows for demolition and felling, p. 22.

20
GREEN MEASURES AND REPLANTING

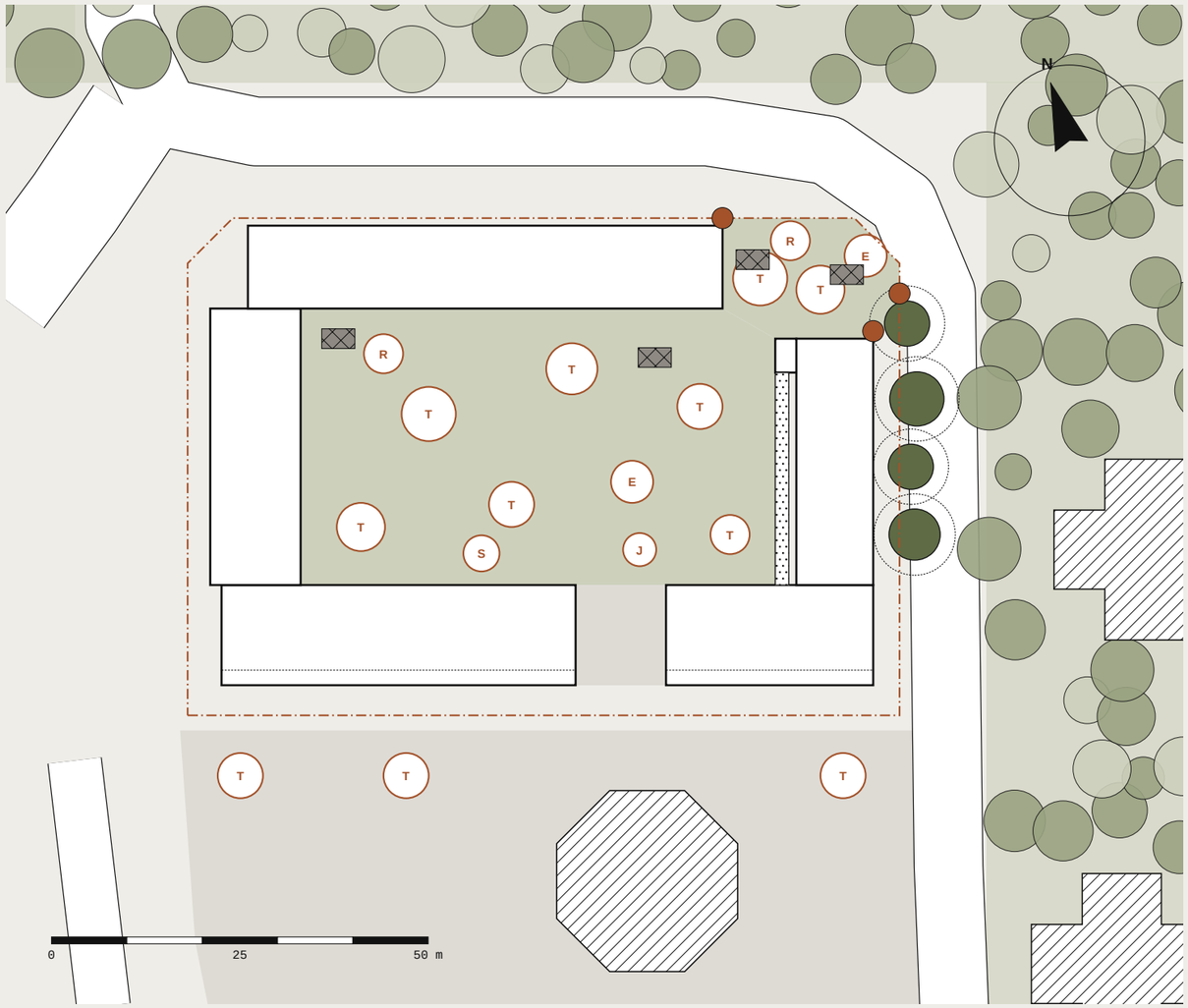
Solna uses the Alnarp model to calculate compensatory planting: the total trunk area of trees removed is compared with that of new trees. The number of trees affected is set after a survey and arborist's report.

The proposal plants 17 trees with 30–35 cm girth: 10 in the courtyard, 4 in the Forest Gate and 3 on the square. Large pines at the forest edge are surrounded by a fenced root protection zone during construction.

Felled trees become habitat piles, high stumps and site-made furniture. Motorborgen's concrete becomes gabion fill and boulders.

TREE PLAN KEY

CORTEN CIRCLE = NEW TREE (T PINE, E OAK, R ROWAN, S WILLOW, J JUNIPER)
GREEN CIRCLE WITH DASHED ZONE = EXISTING TREE + ROOT PROTECTION
GREY BOX = HABITAT PILE · CORTEN DOT = HEDGEHOG PASSAGE



TREE PLAN AND MEASURES 1:800 (A3).

NEST BOXES AND HABITATS

Type	No.	Height	Facing	Location
Swift box	8	+13.3 m	E, N	Eaves, east wing
Bat brick	3	+5.8 m	E	East facade
Bat box	3	4–6 m	S–SE	Trees at forest edge
Tit box Ø 28/32	4	2.5–3.5 m	E–SE	Courtyard
Wood-mould box	2	1–2 m	N–E	Forest Gate
Habitat pile / sand bed	3 / 1	ground	—	Sun and shade
Hedgehog passage	3	13 × 13 cm	—	Fences, walls

CONSTRUCTION PHASE: TIME WINDOWS FOR SPECIES

